Single-Factor Models

Manuel Vilchis Rangel
manuelvilchisrangel@gmail.com

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1 Problem

We are tasked to build a Σ with $N = 35$ based on a one-factor model. $\Sigma = \sigma_m^2 \beta\beta^\top + D$ using:

- $B_i \sim U(0.5, 1.5)$
- $\sigma_m = 0.01$ daily
- Idiosyncratic Vol $\sim U(0.01, 0.02)$

1.1 Explanation

A single-factor model mentions that the returns of an asset have two components

1. **A part that comes from the markets:** When the market goes up 1%, almost every asset tends to go up, some more and some less. The magnitude of the asset's reaction is known as beta (β). An asset with a beta of 1.5 amplifies these movements, whereas one with a beta of 0.5 dampens them.
2. **An idiosyncratic component:** It is unrelated to the market. It is idiosyncratic noise and is assumed to be independent across stocks.

1.2 What information do we obtain for Σ?

Recalling that Σ has variances on the diagonal and covariances in the off-diagonal positions:

1. **Covariance:** Between two distinct variables, the only thing they share is the market, specifically, their reaction to it (β). The covariance of these two variables is the product of their two betas and the market noise:

$$\beta_i \beta_j \sigma_m^2$$

2. **Variance:** The variance of a stock consists of the market component plus the variance of its own noise:

$$\beta_i^2 \sigma_m^2 + d_i^2$$